

PERSONAL INFORMATION

Maria Sameen, Ph.D.



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ABOUT ME

I am a cyber security expert specializing in adversarial machine learning, privacy engineering, and the design of trustworthy digital systems. My work spans model poisoning and phishing detection, privacy-by-design pattern development, and IoT security/privacy labeling — translating technical risk analysis into practical safeguards for both developers and consumers. I investigate deceptive patterns and choice architecture in digital platforms, examining how interface design shapes user behaviour and exposure to harm, with research informing regulatory recommendations (e.g., for Ofcom). My FemTech research audits third-party data practices in health apps, driving calls for stronger GDPR protections. Most recently, I have applied multi-agent LLM systems to industrial control system security, evaluating AI reliability in critical infrastructure defense. My doctoral research exemplifies this interdisciplinary orientation, fusing empirical HCI methods, consumer behaviour analysis, and regulatory theory to address deceptive design as a systemic, cross-stakeholder problem. Holding a Ph.D. in Cyber Security from the Universities of Bristol and Bath, I combine rigorous empirical methods with a consistent focus on translating research into actionable, real-world impact across academia, industry, and policy.

EDUCATION

Sep 2021 – Dec 2025	Ph.D. (Cyber Security)	Uni. of Bristol and Uni. of Bath, UK
Lead Supervisor:	Prof. Awais Rashid	
Co-Supervisor:	Dr. Ryan McConville	
Viva/Defense Examiners:	Prof. Alice Hutchings, University of Cambridge, UK Associate Prof. Luc Rocher, University of Oxford, UK	
Thesis:	“An Empirical Investigation of Deceptive Patterns in Video Games”	
Mar 2019 – Feb 2021	Master of Eng. (Computer Engineering)	Gachon University, South Korea
Mar 2019 - Feb 2020	Master of Sci. (Electronics & Computer Eng.)-Transferred from Hongik University, South Korea	
Supervisor:	Prof. Seong Oun Hwang	
Viva/Defense Examiners:	Prof. Joon S. Lim, Department of Computer Engineering, Gachon University, South Korea Associate Prof. Chang Choi, Department of Computer Engineering, Gachon University, South Korea	
Thesis:	“Machine Learning Approaches for Cyber Security against Phishing and Poisoning Attacks in Artificial Intelligence Environment”	

CGPA: 4.50/4.5

Jan 2015 – Dec 2018 **BS (Computer Science)**

IBA, Karachi, Pakistan

Supervisor: Farhan Ahmed Siddiqui

Thesis: “DigiSikka: Blockchain Technology-based Mobile E-wallets for secure cashless payment system”

CGPA: 3.44/4.0

PROFESSIONAL EXPERIENCE

Apr 2024 – Present **Privacy Patterns Researcher**

University of Bristol and Bristol Energy Network
Beacon House, Queens Road, Bristol, BS8 1QU, UK

Conducting a research study to develop process-level privacy patterns in software designs, validating and evaluating their practical impact with prospective users

Translating empirical findings into concrete design and engineering recommendations/requirements intended for direct adoption by developers; and positioning them as a development tool that engineering teams apply systematically throughout design and implementation

Sep 2021 – Present **IoT Privacy Label Researcher**

REPHRAIN Centre
Beacon House, Queens Road, Bristol, BS8 1QU, UK

Co-creating a security and privacy label for smart-home IoT devices in a joint study with the University of Bath, combining technical risk analysis with user- and expert-facing communication of security guarantees

Co-designing a nutrition-label-style information format intended to make device-level security and privacy properties (data collection practices, update/patch behaviour, known risk factors) legible to non-expert consumers at the point of purchase — addressing the same transparency gap regulators and standards bodies are increasingly targeting for IoT devices

Jan 2026 – Mar 2026 **Research Software Engineer**

England, UK (under NDA)

Designed and built a multi-agent security system (AutoGen) to secure a cyber-physical process, with extensible scaffolding for additional agents and devices

Captured and evaluated agent reasoning across three levels — threat understanding, proposed mitigation, and generated code correctness — to assess multi-agent LLM reliability in detecting and responding to ICS security threats

Instrumented agent coordination via Prometheus/Grafana stack for real-time visibility into agent decision-making

May 2024 – Mar 2025 **FemTech Researcher**

REPHRAIN Centre
Beacon House, Queens Road, Bristol, BS8 1QU, UK

Led a joint study with York University and King's College London examining how women's health data within FemTech apps is shared with third parties, analyzing the most-downloaded menopause applications on the Google Play Store across the UK, and Canada

Cross-referenced app permissions and third-party SDK behavior against each app's published privacy policy and Google Play data-safety declarations, identifying systematic discrepancies — including undisclosed advertising identifiers and event-data sharing inconsistent with stated data-safety claims

Utilized findings to inform and recommend that health data be classified as high-risk under GDPR, that FemTech apps undergo mandatory data protection impact assessments, and that opt-in consent and audit mechanisms be required for third-party data sharing — translating technical privacy-risk analysis into concrete regulatory recommendations

Jan 2023 – Mar 2025 **REPHRAIN Testbed Researcher**

REPHRAIN Centre

Beacon House, Queens Road, Bristol, BS8 1QU, UK

Collaborated with the University of Edinburgh and King's College London to design and build testbed infrastructure for launching virtual machines on abstract network topologies, supporting reproducible privacy and security research at scale — within the same Bristol Cyber Security Group ecosystem that maintains the group's open-source **TestbedOS** virtualization and security-testing platform.

Contributed to the engineering of infrastructure used to evaluate security mechanisms under controlled, configurable network and system conditions.

Jul 2022 – Dec 2022 **Industry Placement**

Ofcom (Office of Communications)

Riverside House, 2a Southwark Bridge Road, London, SE1 9HA, UK

Conducted a joint research study with the Ofcom team on “Choice Architecture and the Safety of Users in Online Platforms,” analyzing how platform design choices (default settings, interface nudges, consent flows) shape user exposure to harm

Assessed how security incidents and user-facing threats are, or could be, identified and managed at scale across major online platforms, producing analysis intended to inform a UK regulator's approach to platform safety oversight

Nov 2021 – Dec 2022 **Testbed Privacy Frameworks Researcher**

REPHRAIN Centre

Beacon House, Queens Road, Bristol, BS8 1QU, UK

Worked with the University of Cambridge to develop a security and privacy measurement framework for a research testbed, supporting systematic identification, documentation, and evaluation of privacy leakage incidents and countermeasures

Defined structured methodology and metrics for detecting and classifying privacy leakage within experimental testbed environments, moving beyond ad hoc incident review toward a repeatable, auditable measurement process

Sep 2020 – Aug 2021 **Graduate Research Student**

Information Security and Machine Learning Lab, Gachon University

1342 Seongnam-daero, Sujeong-gu, Seongnam-si, Gyeonggi-do, South Korea

Conducted applied cyber security research as part of the BK21 FOUR “FAST (Fast, Advanced, Secure and Transaction) Artificial Intelligence Convergence” project, a Ministry of Education (Korea)-funded initiative led by Prof. Seong Oun Hwang spanning 18 foreign academic institutions (including the University of Birmingham, RIT, and Nanyang Technological University) and 18 domestic companies and research institutes

Examined practical security problems within an AI-focused curriculum spanning model robustness, poisoning, and phishing-style attack vectors, contributing technical analysis that fed directly into the lab’s broader AI-security research agenda and international research partnerships

Sep 2018 – Jan 2019 **Software Project Manager**

Pakistan Stock Exchange Limited, Head Office
I.I Chundrigar Rd, Lalazar, Karachi, Karachi City, Sindh, Pakistan

Led the development, stress testing, and implementation of a real-time stock data recovery project for a national financial market infrastructure, coordinating system design, testing cycles, and deployment timelines

Managed cross-functional delivery in a high-availability, mission-critical environment where data integrity and recovery reliability were operationally essential — an early exposure to the engineering discipline required for fault-tolerant, security-sensitive systems

Aug 2018 – Nov 2018 **Research Assistant**

Telecommunication Research Laboratory, IBA
University Rd, University of Karachi, Karachi, Pakistan

Designed and developed an AI-based campus security system that used smartphone sensor data for analytics-driven early-warning detection, identifying patterns associated with suspicious activity and physical harassment

Worked across the full applied research pipeline — sensor data collection, feature design, and detection logic — to translate a real-world physical-safety problem into a working prototype system

Jul 2017 – Aug 2017 **Internship**

State Bank of Pakistan
I.I Chundrigar Rd, Lalazar, Karachi, Karachi City, Sindh, Pakistan

Conducted a research study on the determinants of CPI (Consumer Price Index) inflation using a time-series modeling approach, applying statistical methods to macroeconomic data within a central-bank research setting

Built and evaluated econometric time-series models (e.g. VECM-style specifications) to quantify the relationship between candidate macroeconomic indicators and inflation, an early grounding in rigorous quantitative modeling, hypothesis testing, and data-driven analysis later utilized to predict future trends and inform economic policy

Mar 2017 – May 2017 **Research Analyst**

Stylo Company, Karachi, Pakistan

Conducted a market research study analyzing product sales performance and the key factors driving demand, translating quantitative findings into actionable business insight for the company

Collected and analyzed sales and customer data to identify patterns in purchasing behavior, structuring findings into clear recommendations for business stakeholders for decision-making

Jan 2015 – Mar 2015 **Research Enumerator**

State Bank of Pakistan
I.I Chundrigar Rd, Lalazar, Karachi, Karachi City, Sindh, Pakistan

Conducted structured survey interviews across Pakistan for the national “Consumer Confidence Survey” research project, gathering and analyzing primary data to support macroeconomic sentiment analysis

Administered standardized survey instruments consistently across a geographically dispersed respondent base, ensuring data quality and comparability for downstream statistical analysis at a national scale

PUBLICATIONS

Patent **M. Sameen** et al., *Method and Apparatus for Detecting a Model Poisoning Attack*, (Korean Patent Registration No. 1026268280000), 2024. 01. 15.

Journal Articles Das Chowdhury P, **M. Sameen**, Blessing J, Boucher N, Gardiner J, Burrows T, Anderson R, Rashid A. *Threat models over space and time: A case study of end-to-end-encrypted messaging applications. Software: Practice and Experience.* 2024 May 22.

M. Sameen and S. O. Hwang, "DISTINCT: Data poisoning attacks detection using optimized jaCCard distance," *Computers, Materials & Continua*, vol. 73, no.3, pp. 4559–4576, 2022.

M. Sameen and S. O. Hwang, "TIMPANY—detection of Model Poisoning Attacks using accuracy," in *IEEE Access*, vol. 9, pp. 139415-139425, 2021, doi: 10.1109/ACCESS.2021.3118926.

M. Sameen, K. Han and S. O. Hwang, "PhishHaven—An Efficient Real-Time AI Phishing URLs Detection System," in *IEEE Access*, vol. 8, pp. 83425-83443, 2020, doi: 10.1109/ACCESS.2020.2991403.

Conference Proceedings Zaina D., **M. Sameen**, and Ramokapane, K.M. "Beyond Transparency: IoT Labels as Infrastructures for Trust and Control." In *Proceedings of the 2027 Symposium on Security and Privacy. IEEE, 2027.* (Under Review)

Chitchyan R, Belguith S, Omoronyia I, and **M. Sameen**. "Rethinking Data Collection Practices: User-Centred Patterns for Privacy-Aware (Energy) Services." In *Proceedings of the 2027 International Conference on Software Engineering. IEEE/ACM, 2027.* (Submitted)

Gardiner J, Abaimov S, Williams J, Shahbi F, Anastasakis K, Das Chowdhury P, Ellis W, **M. Sameen**, Samanis E, and Rashid A. " "If You Build it, They will Come" - A Blueprint for ICS-focused Capture-The-Flag Competitions." In *Proceedings of the Sixth Workshop on CPS&IoT Security and Privacy*, pp. 27-40. 2024.

Saad Sohail A, **M. Sameen**, and Ahmed Q. "Latent Dirichlet Allocation Algorithm using Linguistic Analysis." In *2019 International Conference on Green and Human Information Technology (ICGHIT)*, pp. 116-118. IEEE, 2019.

Saad Sohail A, **M. Sameen**, and Ahmed Q. "Understanding Monetary Policy Communication: A VECM Approach Based on Pakistan." In *2019 International Conference on Green and Human Information Technology (ICGHIT)*, pp. 133-138. IEEE, 2019.

Saad Sohail A, **M. Sameen**, and Ahmed Q. "Formal Notations of Linguistic Analysis for Monetary Policy." In *2019 International Conference on Green and Human Information Technology (ICGHIT)*, pp. 119-121. IEEE, 2019.

Magazine Ramokapane, K.M, **M. Sameen**, Zaina D. *Inclusive IoT Privacy Labels. IEEE Security & Privacy Special Issue.* 2024 June 20.

Workshops **M. Sameen** and Rashid A. "Dark Patterns in Video Games: An Exploratory Study." *The Eighth Workshop on Technology and Consumer Protection (ConPro '24)* co-located with the 45th IEEE Symposium on Security and Privacy. 2024.

Das Chowdhury P, **M. Sameen**, Blessing J, Boucher N, Gardiner J, Burrows T, Anderson R, and Rashid A. "Threat Models over Space and Time: A Case Study of E2EE Messaging Applications." *International Workshop on Privacy Engineering (IWPE'23)* co-located with the 8th IEEE European Symposium on Security and Privacy. 2023.

Das Chowdhury P, Blessing J, **M. Sameen**, Anderson R, Gardiner J, and Rashid A. "Towards Human-Centric Endpoint Security." In *International Security Protocols Workshop*. 2023.

Talks **M. Sameen**, and Rashid A. "Dataveillance: An Overlooked Side of Dark Patterns." *Hot Privacy Engineering Techniques Symposium, (HotPETS)*. 2023.

Das Chowdhury P, Gardiner J, **M. Sameen**, Alhindi KM. M, Peden J. G, Ellis W, Mazeli C. A, Tahaei M, and Rashid A. "Building a Privacy Testbed." In *2022 USENIX Conference on Privacy Engineering Practice and Respect*. USENIX Association, 2022.

Posters **M. Sameen**, and Rashid A. "3Ds of Dark Patterns". *ACE-CSR Conference, (NCSC 2022)*.

INDUSTRY ENGAGEMENTS & COLLABORATIONS

Apr 2025 Airbus

Pegasus House, Aerospace Ave, Filton Site, Bristol BS34 7PA, UK

Visited wings design, test, and manufacturing facilities (A320, A330, A350 aircrafts, and A4000M military aircraft); explore Northern Europe's largest concentration of aerospace center specialized in wing design, fuel systems and landing gear; and discussed advanced research programs including Wing of Tomorrow and ZEROe

Mar 2023 BT's innovation labs and Innovation Martlesham

Adastral Park, Martlesham Heath, Ipswich, Suffolk, IP5 3RE, UK

Discussed next generation connectivity platforms with optimized, sustainable networks while exploring the creation of advanced and secured systems through AI and quantum technologies.

Conducted a **Decisions and Disruptions (D-D)**, a tabletop cyber security role playing game, to challenge and train BT team (at Bristol) to manage the security in their critical infrastructure

May 2022 HP Labs

HP Inc Ltd., 2nd Floor, 1 Redcliff Street, Bristol, BS1 6NP, UK

Visited HP's global Security Lab and engaged with HP researchers on current cyber security challenges, exchanging perspectives between academic and industry approaches to applied security research

April 2022 Thales Headquarters

350 Longwater Avenue, Green Park, Reading, RG2 6GF, UK

Discussed and brainstormed real-world security challenges across Defense, Public Security, Civil Aviation, Space, Enterprise, and Cyber security with Thales security professionals and practitioners

Explored security challenges relevant to Thales's identity, data, and application security lines of business, areas where the company has continued to deepen its focus (including, more recently, the shift toward post-quantum cryptography across its identity and communications portfolio)

Contributed an academic, evidence-based lens to discussions typically dominated by commercial and operational security constraints, helping bridge the gap between research-driven threat analysis and deployed industry systems.

TEACHING EXPERIENCE

Sep 2022 – Jan 2024 Graduate Teacher

University of Bristol
Beacon House, Queens Road, Bristol, BS8 1QU, UK

Level: MSc. and Ph.D. Cyber Security

Course: Foundations of Cyber Security

Responsibilities: Designed and facilitated practical security lab sessions; one-on-one mentoring for MSc./Ph.D. cohorts; and supported module assessment design

Jan 2017 – Dec 2017 Teacher Assistant

IBA
University Rd, University of Karachi, Karachi, Pakistan

Level: BS Computer Science

Course: Numerical Analysis, Statistical Inference

Responsibilities: Graded assignments, organized class review sessions, and conducted one-on-one tutorial sessions for BS students

HONOURS AND AWARDS

Sep 2021 – Present IRW (Induction Residential Week) Challenge

REPHRAIN Centre
Beacon House Queens Road Bristol, BS8 1QU, UK

Won a competitive team challenge against three teams — pitching an IoT privacy nutrition label to an industry/academic panel consisted of Dr. Pauline Anthonysamy, Staff Privacy Engineer at Google; a member of the CDT External Advisory Group and; an industry-based Privacy Expert — securing EPSRC project funding from REPHRAIN Centre for Privacy Nutrition Labels research project

Sep 2021 – Sep 2025 UKRI EPSRC CDT Cyber Security (TIPS-at-Scale) Scholarship

University of Bristol and University of Bath
Beacon House Queens Road Bristol, BS8 1QU, UK
Claverton Down, Bath BA2 7AY, UK

Awarded EPSRC (Engineering and Physical Sciences Research Council) CDT (Centre for Doctoral Training) by UKRI (UK Research and Innovation) for Doctorate Degree in Cyber Security

Sep 2021 – Sep 2025 RTSG (Research Training Support Grant)

University of Bristol and University of Bath
 Beacon House Queens Road Bristol, BS8 1QU, UK
 Claverton Down, Bath BA2 7AY, UK
 Granted £2,500 per annum by EPSRC CDT for Doctoral Research in Cyber Security

Mar 2019 – Feb 2021 **Global Scholarship**

Gachon University
 1342 Seongnam-daero, Bokjeong-dong, Sujeong-gu, Seongnam-si, South Korea
 Awarded by Gachon University for Master's Degree in Computer Engineering

Best Paper Awards

- Jun 2020 Gachon University
 1342 Seongnam-daero, Bokjeong-dong, Sujeong-gu, Seongnam-si, South Korea
 Awarded the Best Paper Award and received \$300 as a Token of Appreciation for "Phish-Haven—An Efficient Real-Time AI Phishing URLs Detection System"
- Jan 2019 The 7th International Conference on Green and Human Information Technology
 Kuala Lumpur, Malaysia
 Awarded the Best Paper Award for "Understanding Monetary Policy Communication: A VECM approach based on Pakistan"

Jan 2018 – Dec 2018 **Ignite Grant**

NGIRI (National Grassroot Information and Communication Technologies Research Initiative)
 Ignite-National Technology Fund, Pakistan
 Granted NGIRI funding from Ignite - National Technology Fund for Bachelor's Final Year Project: DigiSikka

Sep 2017 **HEC Laptop**

IBA
 University Rd, University of Karachi, Karachi, Pakistan
 Received a laptop under an HEC (Higher Education Commission) Prime Minister Laptop scheme for securing a high CGPA in 3rd year of BS Degree

Jul 2015 – Sep 2015 **Exchange Participant**

AIESEC
 Denizli, Turkey

- Awarded partial funding from IBA to participate and volunteer in an international social project "Colors of the Universe"
- Represented Pakistan on an international forum comprised of 37 different nationalities
- Participated in cultural and training activities

Jan 2015 – Dec 2018 **State Bank Merit Scholarship**

State Bank of Pakistan
 I.I Chundrigar Rd, Lalazar, Karachi, Karachi City, Sindh, Pakistan
 Awarded by State Bank of Pakistan for BS Degree in Computer Science

CONFERENCES, WORKSHOPS AND BOFS

ACM Conference on Computer and Communications Security

- Oct 2024 The 6th Joint Workshop on CPS & IoT Security and Privacy, Salt Lake City, USA
 Nov 2021 Seoul, South Korea

Privacy Enhancing Technologies Symposium

Jul 2024 Bristol, UK
 Jul 2023 Lausanne, Switzerland

The 45th IEEE Symposium on Security and Privacy

May 2024 The 8th Workshop on Technology and Consumer Protection, San Francisco, USA

Game Developer Conference

Aug 2023 Cologne/Bonn, Germany

Women in Cyber Conference

Aug 2023 Rolls-Royce, Filton, UK

The 8th IEEE European Symposium on Security and Privacy

Jul 2023 International Workshop on Privacy Engineering, Delft, The Netherlands

The 28th International Workshop on Security Protocols

Mar 2023 Cambridge, UK

NCSC ACE-CSR Conference

Jun 2022 Stratford-upon-Avon, UK

Quantum Machine Learning Satellite Workshop

Oct 2019 IBM Korea Quantum Boot Camp
 KAIST, Daejeon, South Korea

- Hands-on learning on IBM Q and Qiskit, open-source Python-based programming frameworks covering all aspects of quantum computing
- Applied Qiskit code for applications in Quantum Chemistry, Machine Learning and Optimization

The 7th Inter. Conference on Green & Human Information Technology

Jan 2019 Kuala Lumpur, Malaysia

PROFESSIONAL ACTIVITIES**Mar 2020 – Present Reviewer**

IEEE

Peer-reviewed research articles for IEEE journals and conferences

Jan 2018 – Mar 2026 Member

Sep 2021 – Mar 2026 Research Member, Bristol Doctoral College, University of Bristol, UK
 Sep 2021 – Dec 2025 Visiting Post-Graduate Researcher, University of Bath, UK
 Jan 2024 – Dec 2024 Professional Member, Association for Computing Machinery (ACM)
 Oct 2018 – Oct 2019 Youth Member, United Nations (Association of Pakistan)
 Jan 2018 – Dec 2018 Member, Institute of Electrical and Electronics Engineers (IEEE)
 Jan 2018 – Dec 2018 Member, Association for Computing Machinery (ACM)

SKILLS**Technical Skills**

Security & Cryptographic Engineering: Threat Modelling, Privacy-by-Design, NIST Cyber security Framework, Security Incident Analysis, Privacy Framework Design

Programming: Java, C/C++, SQL, Python, MATLAB, Intel Assembly, HTML, CSS, Javascript, Rust, Prometheus

Tools: PL/SQL, Eviews, SPSS, R, AWS Blockchain Templates, Grafana

DBMS: Oracle, MySQL

Platforms: Odoo ERP, Oracle E-Business Suite, IBM Hyperledger Composer

Frameworks: Keras, Tensorflow, Pytorch, Autogen (multi-agents)

Operating Systems: Ubuntu, Windows, Kali LINUX

Language Skills

Mother tongue Urdu

Other languages	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken interaction	Spoken production	
English	C2	C2	C2	C2	C2
German	A1	A1	A1	A1	A1

Levels: A1 and A2: Basic user – B1 and B2: Independent user – C1 and C2: Proficient user
[Common European Framework of Reference for Languages](#)

REFERENCES

Dr. Awais Rashid, Lead Ph.D. Supervisor

Designation: Head, Bristol Cyber Security Group (BCSG)

Director, REPHRAIN Centre

Director, EPSRC CDT in Cyber Security: TIPS-at-Scale

PI and Editor-in-Chief, Cyber Security Body of Knowledge (CyBOK)

Professor of Cyber Security, Department of Computer Science

Organization: University of Bristol, UK

Email: awais.rashid@bristol.ac.uk

Dr. Inah Omoronyia, Annual Ph.D. Progress Reviewer

Designation: Programme Director of MSc Cyber Security (Infrastructures Security)

Senior Lecturer in Privacy, Department of Computer Science

Member - Leadership Team of Interdisciplinary CDT

Member - Bristol Cyber Security Group (BCSG)

Researcher, REPHRAIN Centre

Organization: University of Bristol, UK

Email: inah.omoronyia@bristol.ac.uk